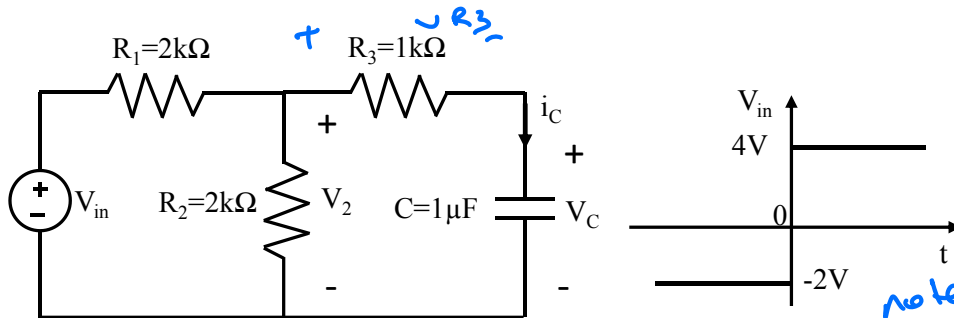


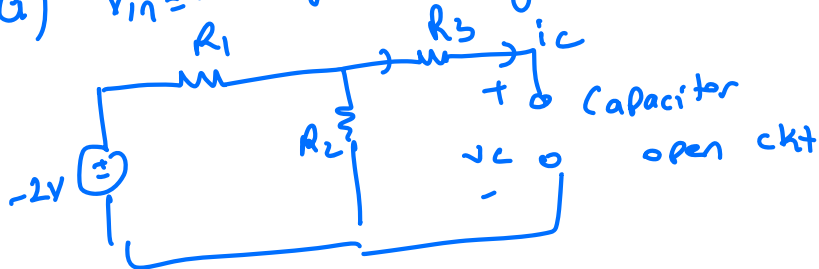
Name:

Student ID:

Duration: 20 minutesFor the circuit given below, $V_{in} = -2V$ for a long time for $t \leq 0$ a) (50 pts) Find and plot $V_C(t)$ for $t \geq 0$ b) (30 pts) Find and plot $i_C(t)$ for $t \geq 0$ c) (20 pts) By using your answer from parts a and b, find and plot $V_2(t)$ for $t \geq 0$ Hint: $V(t) = V_{final} + (V_{init} - V_{final})e^{-t/\tau}$ for a 1st order circuit

note that $i_C = 0$ so no voltage drop across R_3 !

a) $V_{in} = -2V$ for a long time.

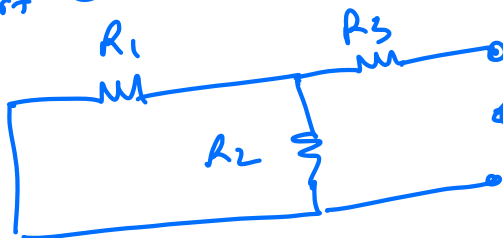


$$V_C(0) = \frac{R_2}{R_1 + R_2} \cdot -2 = -1V$$

Similarly $V_{in} = 4V$

$$V_C(\infty) = \frac{R_2}{R_1 + R_2} \cdot 4V = 2V$$

for Z : short ckt V_{in} .



$$R_{eq} = R_3 + R_1 \parallel R_2$$

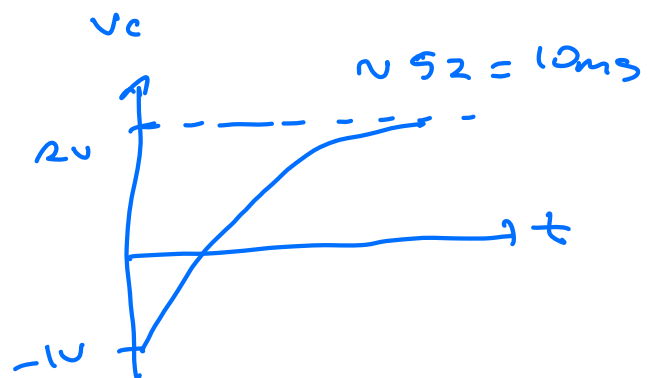
$$R_{eq} = 1 + 2 \parallel 2$$

$$R_{eq} = 2k\Omega$$

$$Z = C \cdot R_{eq}$$

$$Z = 1 \cdot 10^{-6} \cdot 2 \cdot 10^3$$

$$Z = 2ms$$



$$V_C(t) = 2 + (-1 - 2)e^{-t/2} \quad t > 0$$

$$V_C(t) = 2 - 3e^{-t/2} \quad \tau = 2ms$$

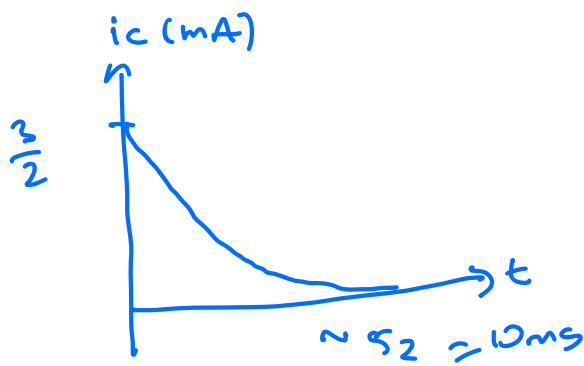
b) $i_C(t) = C \frac{dV_C}{dt}$

$$i_C(t) = C \cdot \frac{d}{dt} (2 - 3e^{-t/2})$$

$$i_C(t) = C \cdot \frac{3}{Z} e^{-t/2}$$

$$i_C(t) = 1 \cdot 10^{-6} \cdot \frac{3}{2 \cdot 10^{-3}} e^{-t/2} \quad \tau = 2ms$$

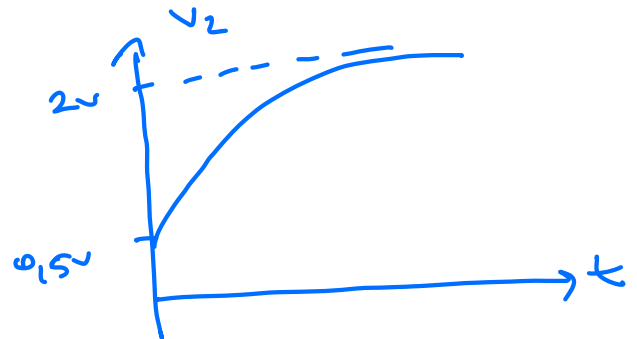
$$i_C(t) = \frac{3}{2} e^{-t/2} \text{ mA} \quad t > 0$$



c) $V_2 = V_{R3} + V_C$

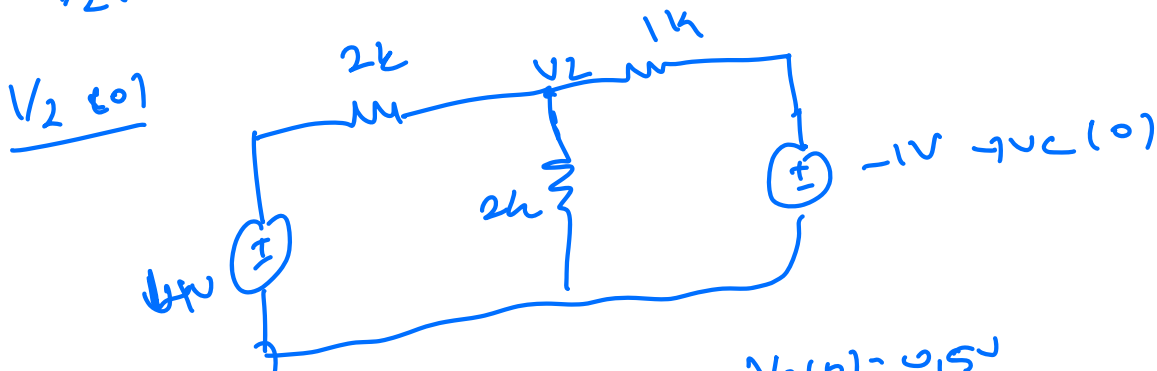
$$V_2(t) = i_c R_3 + V_C = \frac{3}{2} e^{-t/\tau_2} \cdot 1 + 2 - 3e^{-t/\tau_2}$$

$$V_2(t) = 2 - \frac{3}{2} e^{-t/\tau_2} \quad t > 0$$



Let's check by directly calculating V_2 (not required in the solution)

$$V_2(\infty) = V_{CC}(\infty) = 2\text{V}$$



$$V_2(0) = 0.5\text{V}$$

$$V_2(t) = 2 + (0.5 - 2)e^{-t/\tau_2}$$

$$\frac{V_2 - 4}{2} + \frac{V_2}{2} + \frac{V_2 + 1}{1}$$

$$V_2 - 4 + V_2 + 2V_2 + 2 = 0$$

$$4V_2 = 2$$

$$V_2 = 0.5\text{V}$$

$$V_2(t) = 2 - 1.5e^{-t/\tau_2} \quad t > 0$$

same result.